

Movie Recommendation System

32 million movie ratings

(Light gcn)...scalable model...we used it for collaborative filtering (testing purpose)

We used heterogeneous graph attention network...good model for rich dataset

We need to have a model which can give embeddings specifically on the context of movie world only

HGAN /heterogeneous graphSage(no attention):

We reduced tags to around 800(few are DUPLICATE)

Embeddings we got from are mxai model.-1024 (plot +tags)

We made pairs and clusters of movies and cross verified ex : godfather 1 and godfather 2are we getting lot of pairs together ..and checked .

How we collected movies : 10K movies

Do we any metric/list to get the popular movies ??

Imdb – top 250

The metric imdb uses is Bayesian

We used the formula

Imdb provides data of all the movies until now (in csv), we used formulas on these movies and shortlisted 10k movies.(all languages probably).

We used two scores : rating score and popularity score and applied the Bayesian formula AND SORTED IT OUT to get 10k movies.

Prblm : few movies have high rating but by very less number of users (biased movie and users).

We can know the total no of movies in each genre

Distribution of movies on genre and take the top ceiling

Final approach: we decided to have 250 genres for each movie

From there 8,000 are collected and 2000 from the Bayesian formula

How the strategy helped us to remove those movies which have higher rating because of biasness?????????

We need to get information from the movies:

Tags, plot summary, genre, collections, directors, year of release, ratings....

We need to get this info from imdb and tmdb

We used selenium for html scrapping?

it automates the browser ..it takes the plot summary from imdb and does in such a way that we don't get rate limit.

We have to build a knowledge graph with the data: need to decide nodes and edges

Nodes can have static values and trainable values.

For semantic anchors: we got the list and we gave embeddings regarding their top movies

For genre: here we define what each genre is and give embeddings for that. (19 genres from tmdb/imdb).

For collection: embeddings will be mean of embeddings of all movies in that collection.

We trained the model using hgan, and tried to predict the scores of the tags as a proxy task ...so we can get better embeddings for the movie.

Faulty approach:

We had two tasks a. better clustering, collaborative filtering in the starting...using the same loss function and it is conflicting in nature ...so we removed

Now we are using EASE for collaborative filtering :

$X - XW$

Here X : user to movie matrix

W : MOVIE TO MOVIE MATRIX

Rich dataset: